# TP3 : Filtres actifs

## Groupe 2 LT2B :


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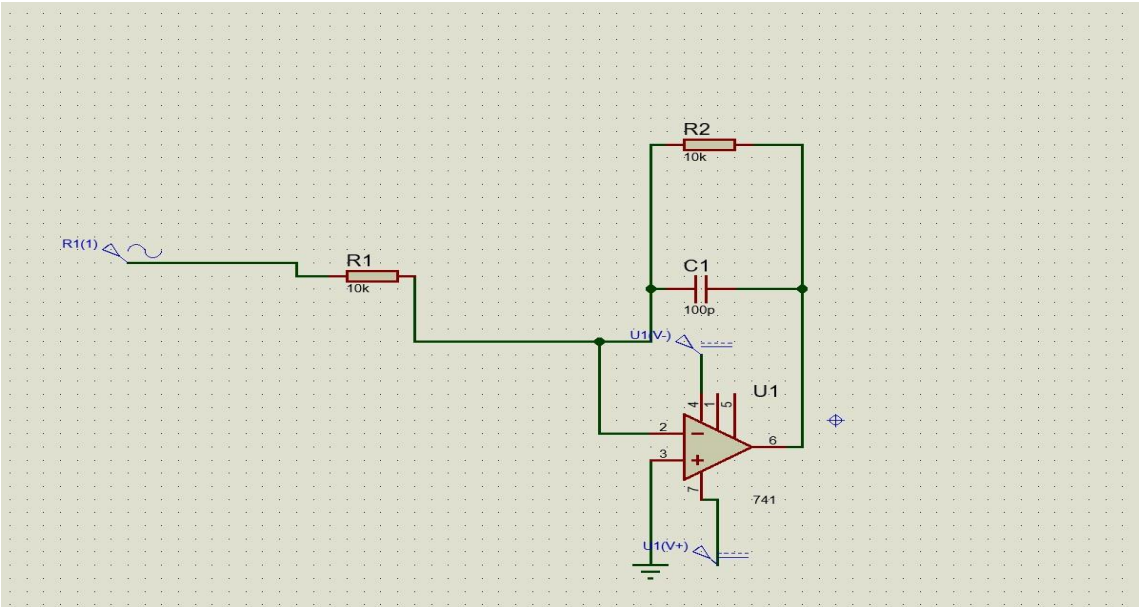
-Sellami arij

-Marouani malek


## Partie A :

### Le montage sur isis :




1)-L'aop est ideal

Donc{v+=v-/i+=i-=0}

Ona v+=0=>d'apres le circuit

$$v- = \frac{\frac{ve}{R1}+\frac{vs}{z1}}{\frac{1}{R1}+\frac{1}{z1}}$$ avec Z1=$\frac{R2}{R2jcw+1}$

donc v-=$\frac{ve*z1+vs*R1}{z1+R1}=\frac{\frac{ve*R2}{R2jcw+1}+vs*R1}{\frac{R2}{R2jcw+1}+R1}=\frac{ve*R2+vs*R1*R2jcw+vsR1}{R2+R1R2jcw+R1}$

v+=v-

ve*R2+vs*R1*R2jcw+vsR1=0

ve*R2+vs(R1R2jcw+R1)=0

ve*R2=-vs*(R1R2jcw+R1)

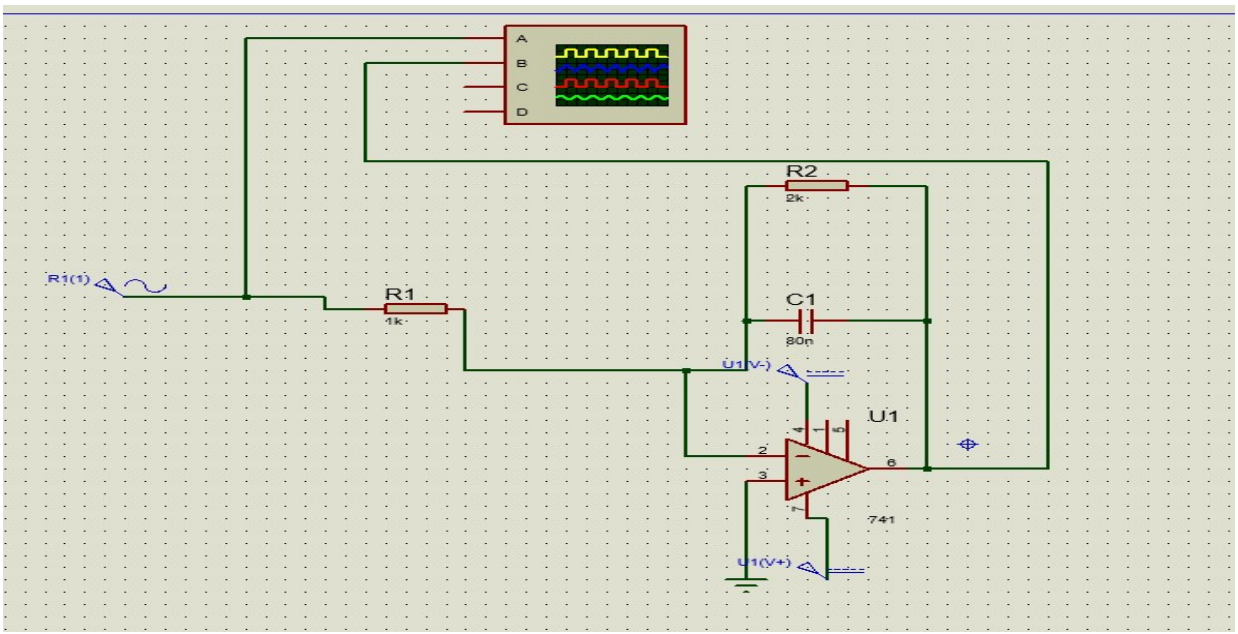
$$\frac{Vs}{Ve}=\frac{-R2}{R1*R2jcw+R1}=\frac{-R2}{R1(R2jcw+1)}=\frac{-R2}{R1}*\frac{1}{R2jcw+1}=H(jw)$$

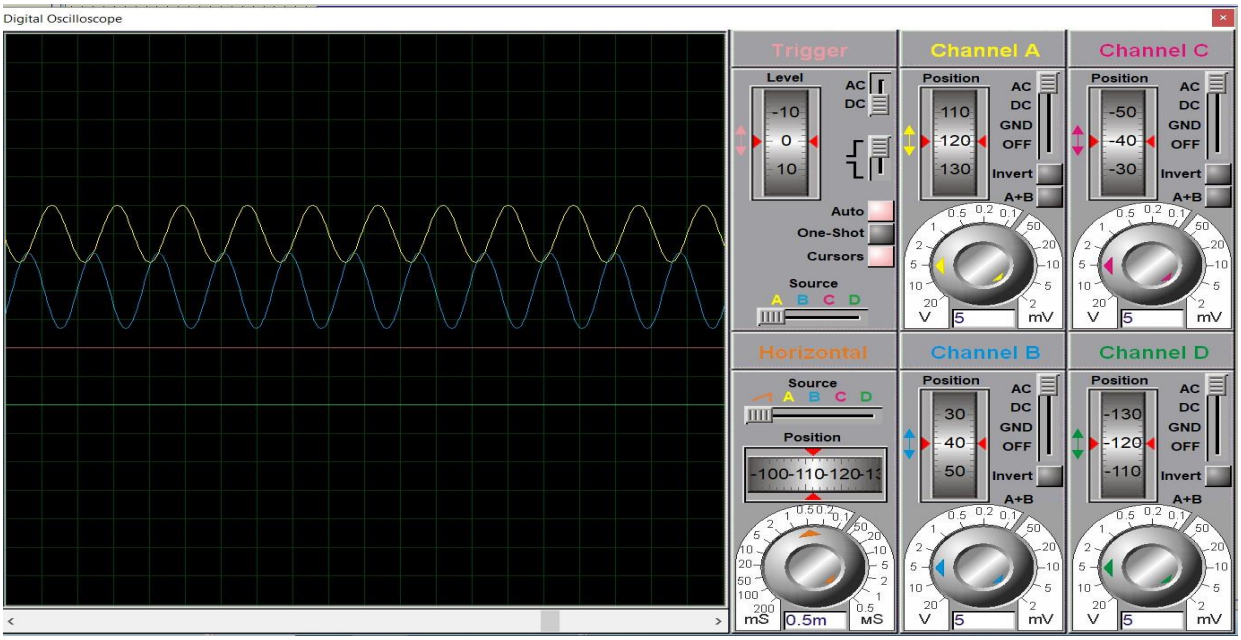
2)-fc=1/2*pi*R2*c=1/2*pi*2*10^3*80*10*^-9=995.22HZ

3)-

Ve=5v ;G(db)=20log(Vs/Ve) ;ΔΦ=(2*pi*ΔT)/T

| F(hz) | 700 | 800 | 900 | 995 | 1000 | 1200 | 1500 | 2000 | 2500 | 3500 | 4500 | 6000 | 8000 | 10000 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Φ | 3pi/4 | Pi/2 | Pi/2 | Pi/2 | 2*pi/3 | 2*pi/3 | 2*pi/3 | 2*pi/3 | 2*pi/3 | 2*pi/3 | 2*pi/3 | 2*pi/3 | 2*pi/3 | 2*pi/3 |
| Vs | 8.13 | 7.75 | 7.38 | 7.13 | 7 | 6.25 | 5.50 | 4.38 | 3.50 | 2.75 | 2.13 | 1.63 | 1.25 | 1 |
| Vs/Ve | 1.62 | 1.55 | 1.47 | 1.42 | 1.4 | 1.25 | 1.1 | 0.87 | 0.7 | 0.55 | 0.42 | 0.32 | 0.25 | 0.2 |
| G(db) | 4.19 | 3.80 | 3.34 | 3.04 | 2.92 | 1.93 | 0.82 | -1.20 | -3 | -5.1 | -7.5 | -9.8 | -12 | -14 |






## 4)Diagramme de Bode

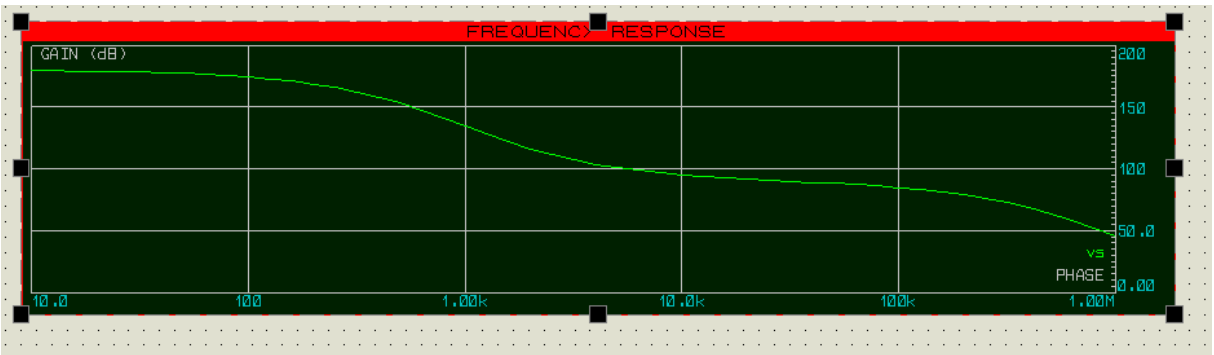



Diagramme de phase

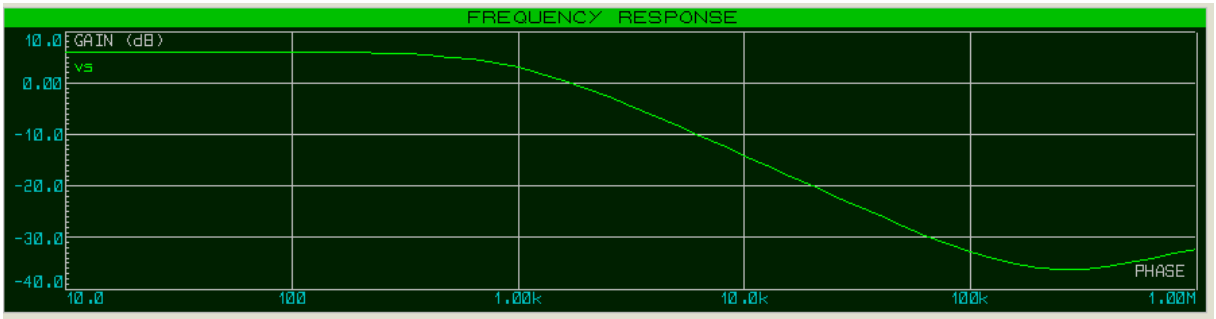
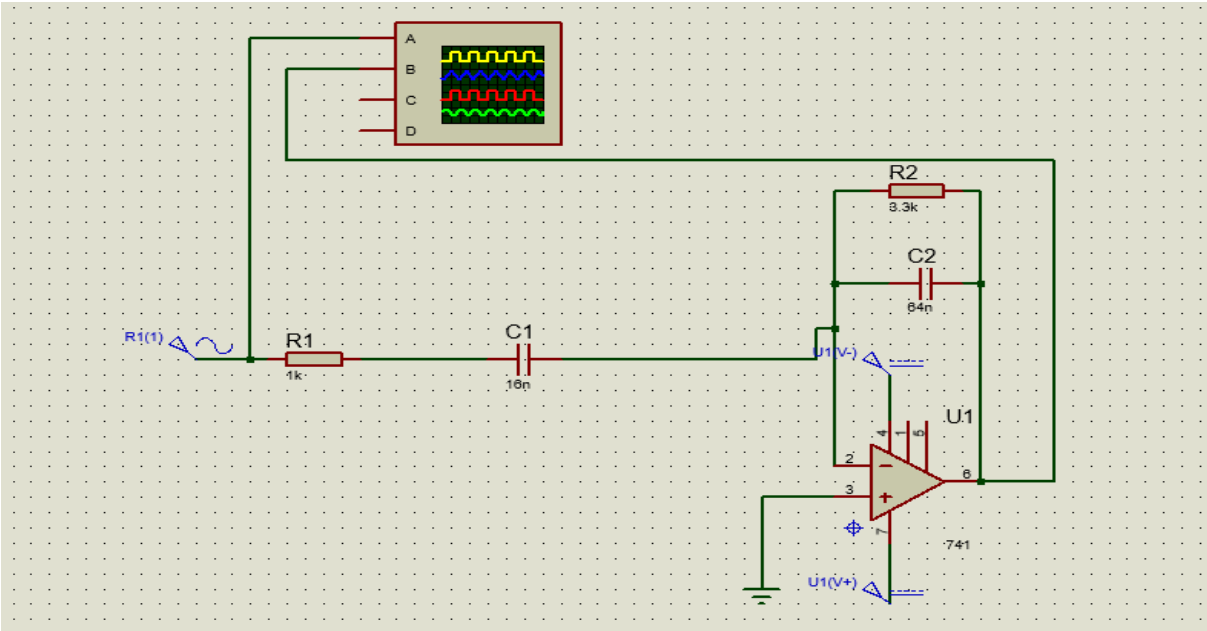



Diagramme de gain(db)

# Partie B :

Schema sur isis :




1) L'aop est ideal

Donc{v+=v-/i+=i-=0}

Ona v+=0=>d'apres le circuit

$$v-=\frac{\frac{ve}{z1}+\frac{vs}{z2}}{\frac{1}{z1}+\frac{1}{z2}}$$

avec z1=R1+zc1=R1+$\frac{1}{jc1w}$=$\frac{R1jc1w+1}{jc1w}$

$$Z2=R2//Zc2=\frac{R2*Zc2}{R2+Zc2}=\frac{R2}{R2jc2w+1}$$

Et ona V+=V-

Donc $\frac{ve}{z1}+\frac{vs}{z2}$=0

$$=>\frac{ve}{z1}=-\frac{vs}{z2}$$

$$=>\frac{vs}{ve}=\frac{-z2}{z1}=>\frac{-R2}{R2jc2w+1}*\frac{jc1w}{R1jc1w+1}=\frac{-R2}{R1}*\frac{1}{jR2c2w+1}*\frac{jc1R1w}{R1jc1w+1}=H(jw)$$

2)-w=2πf

Donc R2c2w=R2c2*2πf=$\frac{f}{fc2}$=> fc2=$\frac{1}{2piR2c2}$=$\frac{1}{2*pi*3.3*10^3*64*10^{-9}}$ =753.57Hz

R1c1w=R1c1*2πf=$\frac{f}{fc1}$=> fc1=$\frac{1}{2piR1c1}$=$\frac{1}{2*pi*1*10^3*16*10^{-9}}$ =9947.18Hz

3)-

| F(hz) | 100 | 500 | 1000 | 1500 | 2000 | 3000 | 5000 | 7000 | 8500 | 10000 |
|---|---|---|---|---|---|---|---|---|---|---|
| Vs | 625*10*-3 | 2.75 | 4 | 4.25 | 4.63 | 4.63 | 4.68 | 4 | 3.75 | 3.5 |
| Vs/Ve | 0.03 | 0.14 | 0.2 | 0.212 | 0.23 | 0.23 | 0.234 | 0.2 | 0.18 | 0.175 |
| G(db) | -30.45 | -17.07 | -13.97 | -13.47 | -12.76 | -12.76 | -12.61 | -13.97 | -14.89 | -15.13 |

4)-

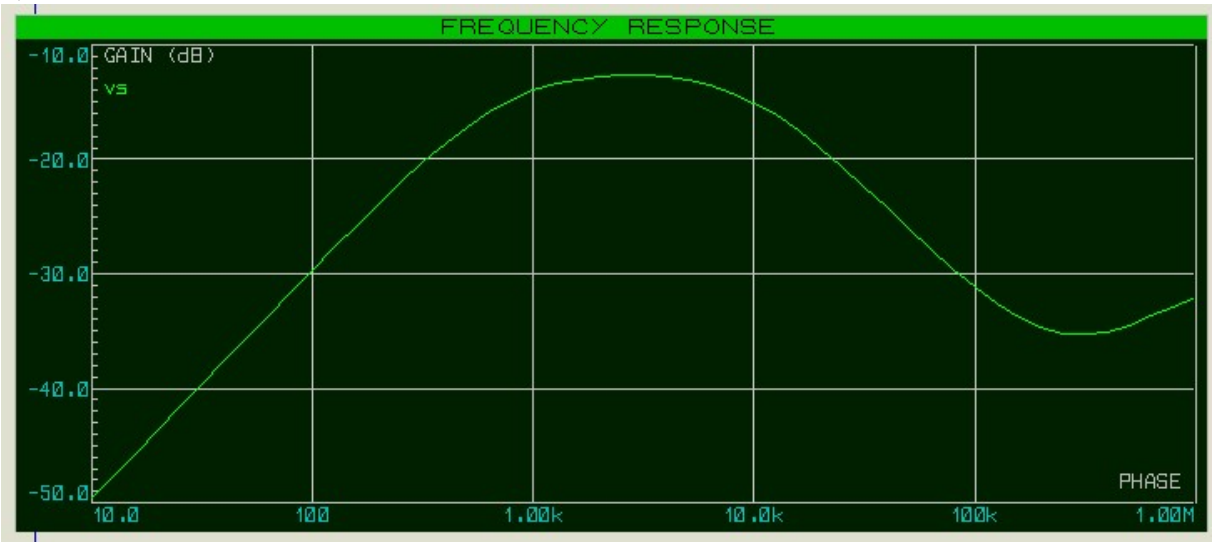



Diagramme d'amplitude

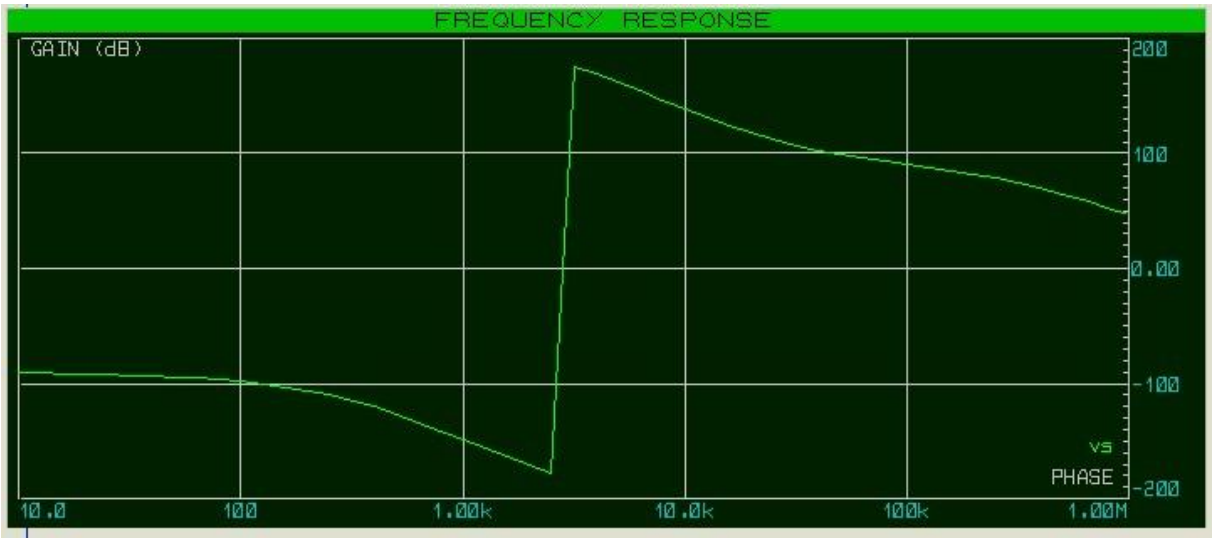
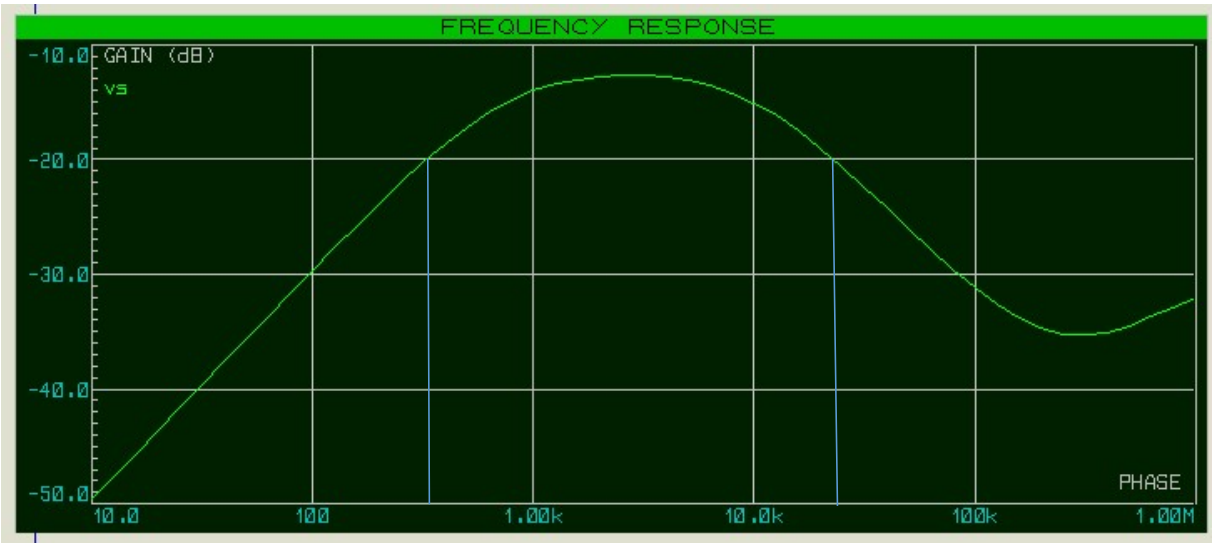



Diagramme de phase

5)-




Wc1 wc2

Donc la bande passante est wc2-wc1

6)-

Si R1=R2=2kr ;C1=C2=160nF ;

Le schéma et le diagramme

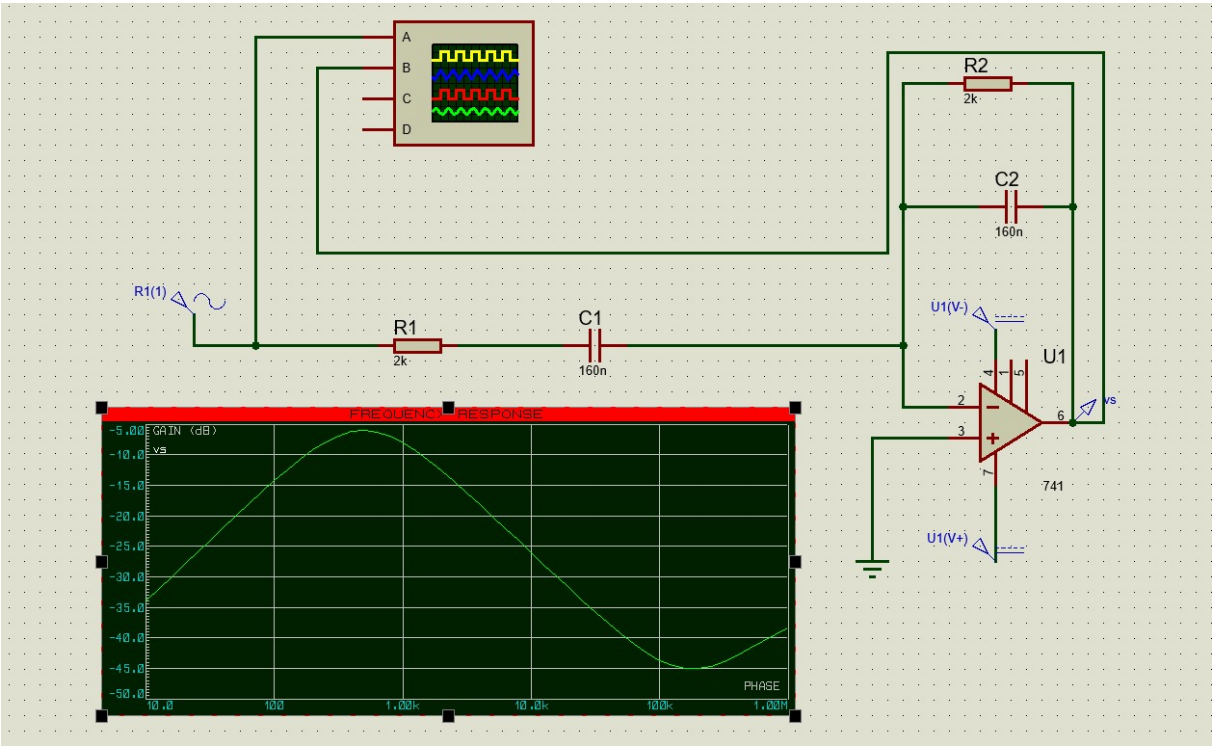
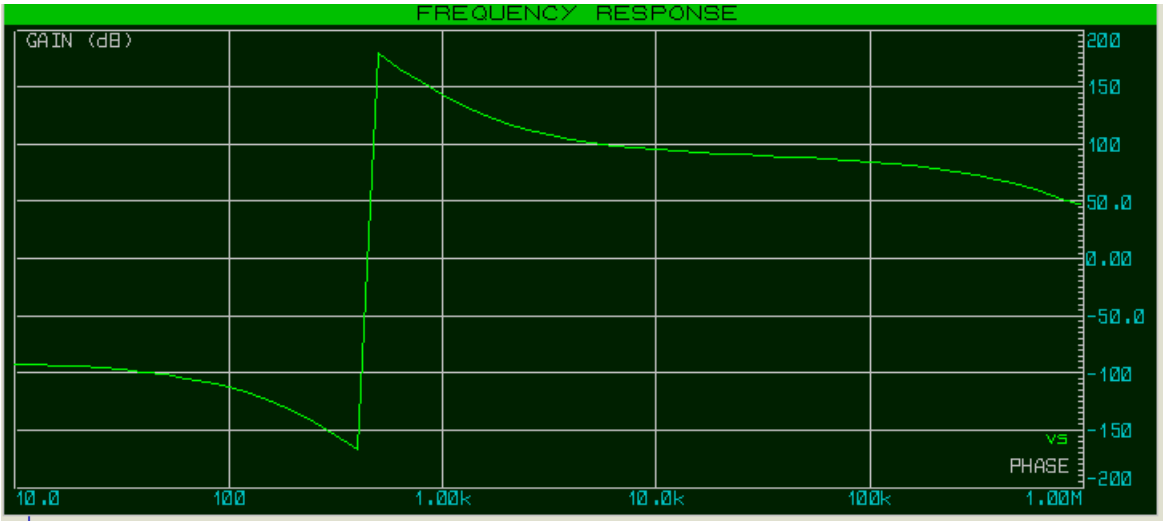



Diagramme de phase :


FREQUENCY RESPONSE
GAIN (dB)
200
150
100
50.0
0.00
-50.0
-100
-150
-200
vs
PHASE
10.0
100
1.00k
10.0k
100k
1.00M